Received: 5 November 2019 | Revised: 24 March 2020 | Accepted: 28 March 2020

DOI: 10.1111/nmo.13858

ORIGINAL ARTICLE

Neurogastroenterology & Motility NGM WILEY

Purgative/laxative actions of *Globularia alypum* aqueous extract on gastrointestinal-physiological function and against loperamide-induced constipation coupled to oxidative stress and inflammation in rats

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Abstract

Background: Chronic constipation is a gastrointestinal functional disorder which affects patient quality of life. Therefore, many studies were oriented to search herbal laxative agents. In this study, we investigated the effect of *Globularia alypum* L. leaves aqueous extract (GAAE) against loperamide (LOP)-produced constipation.

Methods: Animals were given LOP (3 mg/kg, *b.w.*, *i.p.*) and GAAE (100, 200, and 400 mg/kg, *b.w.*, *p.o.*) or yohimbine (2 mg/kg, *b.w.*, *i.p.*), simultaneously, for 1 week. Gastric-emptying test and intestinal transit were determined. Colon histology was examined, and oxidative status was evaluated using biochemical-colorimetric methods.

Key Results: GAAE ameliorates significantly gastric emptying (64% to 76.5%) and intestinal transit (66.65% to 84.73%). LOP negatively influenced defecation parameters and generated a stress situation. GAAE administration in contrast ameliorated those parameters and re-established oxidative balance.

Conclusion: GAAE showed a modest action against oxidative stress and decreased LOP effect and thereby can be considered a pharmacological agent in constipation.

KEYWORDS

constipation, *Globularia alypum* L., herbal laxative agents, oxidative stress, rat, stomach/gut movements

1 | INTRODUCTION

Constipation is a functional defecation disorder affecting up to 20% of the general population.[1] It is characterized by infrequent bowel movements and difficulty in passing stools.[2] Constipation affects patient life quality and can cause many disruptions such as significant psychological distress, abdominal distension, vomiting, gut obstruction, and perforation.[3] Therefore, patients suffering from constipation are at higher risk to colonic cancer, hemorrhoids, and other gastrointestinal illnesses.[3] Constipation can be the result of insufficient fiber and fluid intake or side effects of medication.[4] The intestinal mucosal barrier is covered by hydrate gel secreted by epithelial goblet cells which protect against mechanical damage and chemical irritants in the lumen.[5] The intact epithelial cell layer controls absorption of nutriments and allows passive transport of ions and water by the apical junctional complex. Direct epithelial cell damage results in a marked loss of barrier function.[6]

Abbreviations: Ca^{2+}, calcium; GAAE, *Globularia alypum* leaves aqueous extract; GPx, glutathione peroxidase; H_2O_2, hydrogen peroxide; LOP, loperamide; MDA, malondialdehyde; OH^-, hydroxyl anion; ROS, Reactive oxygen species; -SH, sulfhydryl groups; SOD, superoxide dismutase.

Many drugs are available to treat constipation such as senna, correctol, exlax, and senokot. However their use is limited by high cost and side effects.[7] The majority of constipation drugs regulate the motility of the gastrointestinal tract. Cisapride was the first generation promotility agent.[8] The use of agents which act as a selective type 2 chloride channel (ClC-2) activator in the apical membrane of the gastrointestinal epithelium can be an alternative approach to constipation treatment. ClC-2 increases intestinal chloride secretion and thereby intraluminal fluid collection in the gut which facilitates transit in the intestine and eases stool passage.[9]

Globularia alypum L. (GA), a plant belonging to the *Globulariaceae* family, has been used traditionally by diabetics and in different pathologies. The analyses of the aqueous extract of GA leaves showed its abundance in secondary metabolites including flavonoids, iridoids, and phenylethanoid glucosoides.[10] GA extract showed many in vivo and in vitro biological activities such as anti-inflammatory activity against acetic acid-induced colitis in rats,[10] antioxidant effect against carrageenan-induced paw-edema in mice,[11] in vitro antimicrobial activity,[12] spermatogenesis activation in rats,[13] and vasodilatory effect in rat mesenteric arterial bed.[14]

In the present study, we investigate the purgative/laxative effects of an aqueous extract of GA (GAAE) on gastrointestinal-physiological functions and against loperamide (LOP)-induced constipation coupled to oxidative stress in rats.

Key Points

- *Globularia alypum* L. aqueous extract (GAAE) has improved a laxative product against constipation on rats. This research allows the use of this plant as a natural remedy instead of chemical drugs.
- Constipation induced by loperamide on 60 rats followed by histological and biochemical tests were done to evaluate the effect on oxidative stress and inflammation.
- GAAE showed an acceleration of gastric emptying and gastrointestinal transit *in vivo*. Morevoer, GAAE has modulated the oxidative stress by antioxidant enzyme regulation and decreased colon inflammation

2 | MATERIALS AND METHODS

2.1 | Reagents

Yohimbine, loperamide, red phenol, gum Arabic, 55-dithiobis (2-nitrobenzoic acid) (DTNB), Trichloroacetic acid (TCA), Butylhydroxytoluene (BHT), methanol, charcoal meal, ether, Bovin serum albumin (BSA), NaCl, Bovin catalase, Epinephrine, Chlohydric acid (Hcl), GSH, tris, tartrate, methyl cellulose, ethylene tetra-acetic (EDTA), sodium hydroxide (NaOH), folin-ciocalteu, salicylic acid, copper sulfate (CuSO4), thio-barbituric acid (TBA), sodium carbonate (Na_2CO_3), sodium bicarbonate ($NaHCO_3$), and hydrogen peroxide (H_2O_2) were purchased from (Biomaghreb, Ariana, Tunisia) and were ISO 9001 certified.

Ketamine, monopotassium phosphate (KH_2PO_4), sodium phosphate dibasic dihydrate (Na_2HPO), and iron sulfate ($FeSO_4$) were purchased from (Sigma-Aldrich).

2.2 | Plant collection and extract preparation

GA was collected in March 2017 from the mountains area of Bou Salem city (Jendouba Governorate, North West of Tunisia). A voucher specimen was deposited in the herbarium of the Higher Institute of Biotechnology of Beja, Jendouba University, with the number GA-ISBB-02/04/17. The GA leaves were recuperated and dried in an incubator at 40°C during 72 hours then powdered in an electric blender (MOULINEX Ovatio 2). The powder obtained was dissolved in bi-distilled water and incubated for 24 hours under agitation by a shaking bath. After that the extract was centrifuged for 10 minutes at 10 000 *g* and the supernatant was finally lyophilized and stored at −80°C until use.

2.3 | Animals

Adult male Wistar rats (weight 250 ± 30 g, 17 weeks old, housed 5 per cage) were purchased from the Pasteur Institute of Tunis (Tunisia) and used following the local ethics committee of Tunis University of the use and care of animals and in accordance with the NIH recommendation. Animals were housed in communal cages (12/12 hours light-dark cycle, temperature 22 ± 2°C, provided with standard food (Table 1) and water available ad libitum.

2.4 | Induction of constipation in rats

Animals were divided into 6 equal groups with ten in each. The first group was a negative control receiving physiological solution (0.9% NaCl; 5 mL/kg, p.o.). The other groups received LOP (3 mg/

TABLE 1 Composition of standard diet provided by El-Badr Society (Utique, Tunisia)

Lipid % Energetic contribution	03 5.22
Carbohydrate % Energetic contribution	40 69.56
Protein % Energetic contribution	14.5 25.22
Iron	50 mg/kg
Calcium	1.45 mg/kg
Energetic value	11.40 kj/g

kg, b.w.) by intra-peritoneal injection twice daily (at 9 hours and 17 hours). After 2 hours, three of them were treated with different concentrations of GAAE (100, 200 and 400 mg/kg, b.w.) and one with yohimbine, a drug which facilities colon transit (2 mg/kg, b.w. i.p.). The same treatment was repeated for 5 days. At the end of the experiment, all animals were sacrificed after anesthesia with ketamine (70 mg/kg, b.w. i.p.) and colons were recuperated. After cleaning with physiological solution, colons were homogenized in phosphate buffer saline. Samples were then centrifuged at 3000 *g* for 15 minutes at 4°C, and supernatants were used for biochemical assays.

2.5 | Body weight, food intake, water consumption, and fecal parameter measurements

During the experiment and daily, rat weight, water, and food intake were measured. Every day one hour after extract and drug administration, the excreted fecal pellets were collected directly during 4 hours. Their fresh and dry weight was measured to determine fecal water content. Excreted fecal pellets during 24 hours were also collected determining their total number.

Total fecal pellets collected and the quantity of cinder were determined according to AFNOR.[15] The cinder content corresponds to the residual mass obtained after incineration of the pellets in a muffle furnace at 550°C for 5 hours.

The water content was calculated as follows:

$$\text{Fecal water content}(\%) = [(\text{fecal wet weight} - \text{fecal dry weight})/\text{fecal wet weight}] \times 100.$$

The mineral content was calculated as follows:

$$\text{Mineral content}(\%) = [(\text{fecal dry weight} - \text{cinder weight})/\text{fecal dry weight}] \times 100.$$

2.6 | Gastrointestinal transit

The gastrointestinal transit was evaluated by the charcoal meal test described by Ali and Bashir.[16] Therefore, the rats were divided into 5 groups of 6 rats each and starved for 16 hours before the experiment. The first group received saline solution (0.9% NaCl, 5 mL/kg, p.o.) and was used as negative control. Three groups were treated orally with GAAE (100, 200, and 400 mg/kg, b.w). The latter group received yohimbine (2 mg/kg, b.w. i.p.) representing the positive control. After two hours, all groups received by gavage standard charcoal meal (10% charcoal in 5% gum Arabic, 1 mL/rat). All rats were sacrificed after 30 minutes, and the small intestine was carefully removed (pylorus region to cecum). The distance traveled by the charcoal and intestine length were both measured.

$$\text{Gastrointestinal Transit} = \text{Distance traveled by charcoal(cm)}/\text{intestinal lenght(cm)} \times 100.$$

2.7 | Gastric emptying

The gastric emptying rate was determined by the red phenol method.[17] Briefly, five groups of rats of 6 animals each were used. Two groups were used as positive and negative controls. One of them received physiological solution (0.9% NaCl, 5 mL/kg, p.o.), and the second was treated with yohimbine (2 mg/kg, b.w. p.i.). The other three groups were treated orally with three concentrations of GAAE (100, 200, and 400 mg/kg, b.w.). A test meal (1.5 mL/rat) formed by 50 mg phenol red in 100 mL aqueous methyl cellulose (1.5%) was orally administered after 1 hour. A supplementary animal group received just the test meal and sacrificed immediately using like standard (Using standard mode). The other animals were sacrificed 1 hour after that. The stomach was recuperated and homogenized with its content in 100 mL of NaOH (0.1 N) and left to settle for 1 hour. Five milliliter of supernatant were added to 0.5 mL of trichloroacetic acid (20%) and centrifuged 20 minutes at 1800 g. Four milliliter of NaOH (0.5 N) were added to the supernatant. Sample absorbance was determined at 560 nanometer (nm) wavelengths.

$$\text{Gastric emptying rate}(\%) = (1 - \text{absorbance of treated}/\text{absorbance of standard}) \times 100.$$

2.8 | Histopathological study

After sacrifice, samples from each treated group were recuperated and fixed in a 10% paraformaldehyde for histopathological study. Colon tissues were embedded in paraffin, sliced into 5 µm sections, deparaffinized, and rehydrated. These sections were stained with hematoxylin and eosin (H & E) according to standard histological procedures and were examined for their morphology. For the evaluation of the mucosal layer thickness and mucus-producing cell number sections, sections fixed in 10% paraformaldehyde were deparaffinized with xylene and rehydrated. Subsequently, sections were rinsed with distilled water and stained with Alcian blue.

2.9 | Biochemical assays

2.9.1 | Antioxidant enzyme activities assays

The dismutase superoxide (SOD) activity was determined following Misra and Fridovich's method.[18] In basic pH, the SOD capacity to transform superoxide anion to H2O2 is evaluated in competition with autoxidation of epinephrine on adenochrome by superoxide anion. Therefore, sample was added to the mixture of bovine catalase (0.4 U/µL), epinephrine (5 mg/mL) and sodium carbonate/bicarbonate (62.5 mmol/L; pH 10.2). The SOD activity was determined by detecting absorbance changes at 480 nm.

The Glutathione Peroxidase (GPx) activity was determined according to Floché and Günzler's method.[19] The reaction medium contained 0.1 mol/L phosphate buffer (pH 7.4), 4 mmol/L GSH, 5 mmol/L H2O2 with supernatant colon sample. After one-minute incubation, the reaction was stopped by addition of 5% TCA. The

TABLE 2 GAAE and yohimbine restore food intake and water consumption affected during LOP-induced constipation

	Food intake (g/day)	Water consumption (mL/d)
Control	14.27 ± 0.32	23.9 ± 0.6
LOP	5.47 ± 0.43*	20.2 ± 1.04
LOP + GAAE 100	7.54 ± 0.43#	21.25 ± 1.4
LOP + GAAE 200	8.95 ± 0.59#	21.45 ± 0.79
LOP + GAAE 400	10.74 ± 0.38#	21.7 ± 0.55
LOP + Yohimbine	12.1 ± 0.27#	22.29 ± 1.04

Note: Animals received LOP (3 mg/kg, b.w., i.p.) then treated by various GAAE concentrations (100, 200, and 400 mg/kg, b.w., p.o.), or yohimbine (2 mg/kg, b.w., i.p.) or NaCl (0.9%) over 5 d. The data are expressed as means ± SEM (n = 10).

*$P < .05$ compared to the control group and #$P < .05$ compared to LOP group (ANOVA test).

mixture was centrifuged (1500 *g*, 5 minutes), and then, phosphate buffer followed by 10 mmol/L DTNB was added to supernatant. The absorbance was measured at 412 nm.

2.9.2 | Lipid peroxydation measurement

Lipoperoxidation level was evaluated via malondialdehyde 12 measurement test.[20]20 Briefly, diluted samples from colons were added to TCA-BHT solution (1% BHT, 20% TCA) then centrifuged 1000 *g* for 5 minutes. Tris-TBA (26 mmol/L tris, 120 mmol/L TBA) and 0.6 mol/L HCl were incubated with supernatant at 80°C during 10 minutes. After cooling, the absorbance of different samples was spectrophotometrically measured at 532 nm. The extinction coefficient for MDA-TBA complex (1.56 × 105 per mol/L/cm) was used to determine MDA levels.

2.9.3 | Sulfhydryl groups (-SH) measurement

Determination -SH was evaluated according to Ellman's method.[21] The experiment is based on the absorbance difference at 412 nm after photometric reaction between –SH and 5,5'-dithiobis-(2nitrobenzoic acid) (DTNB). Briefly, colon supernatants were mixed with 100 µL of SDS (10%) and 800 µL of phosphate buffer (10 mmol/L, pH 8) given the first optical density measurement. Then, 100 µL of DTNB were added and the mix was incubated 60 minutes at 37°C given the second optical density measurement. Results were calculated using a molar extinction coefficient of 13.6 × 103 per mol/L/cm.

2.9.4 | Free radicals measurement

H_2O_2 level in treated colon was determined following Dingeon's method.[22] In the presence of peroxidase, The H2O2 reacts with p-hydroxybenzoic acid and 4-aminoantipyrine giving rise to a pink color which refers to quinoneimine. The reaction is quantified by absorbance at 505 nm.

Hydroxyl anion (OH−) level was determined by Payá et al method.[23] Briefly, supernatants from colon samples were incubated with Fe SO4 (9 mmol/L), H2O2 (8.8 mmol/L), and salicylic acid (9 mmol/L) for 1 hour at 37°C. Absorbance was recorded at 510 nm.

2.9.5 | Calcium measurement

Calcium (Ca^{2+}) levels in treated colons were determined according Stern and Lewis method.[24] Ca^{2+} which precipitates as Ca^{2+} oxalate forms a complex with the *o*-cresolphthalein and the reaction is colorimetrically measured at 570 nm.

2.9.6 | Protein determination

Measurement of protein concentration of each colonic sample was determined using Lowry's methods modified by Hartree.[25] The method is based on the reaction of copper iron produced by peptide bonds oxidation with folin-ciocalteu reagent under alkaline conditions. Reduced Folin is measured at 660 nm, and protein determination is carried out using the bovine serum albumin (BSA) as standard.

TABLE 3 Positive influence of GAAE and yohimbine on food fecal parameters after perturbations caused by LOP administration

Groups	Fecal number (n/24h)	Fecal stimulation (%)	Fecal water content (%)	Mineral content (%)
Control	17.2 ± 0.92	–	53.6	22.23
LOP	7.25 ± 0.88*	42.15	29.5*	16.84*
LOP + GAAE 100	9.75 ± 0.51#	56.68	39.6#	18.08#
LOP + GAAE 200	11.37 ± 0.59#	66.1	42.72#	19.34#
LOP + GAAE 400	13.45 ± 0.58#	78.19	47.6#	19.78#
LOP + Yohimbine	14.62 ± 0.62#	85	46.3#	19.78#

Note: Animals received LOP (3 mg/kg, b.w., i.p.) then treated orally by various GAAE concentrations (100, 200, and 400 mg/kg, b.w., p.o.), or yohimbine (2 mg/kg, b.w., i.p.) or NaCl (0.9%) over 5 d. Data are expressed as means ± SEM (n = 10).

*$P < .05$ compared with the control group and #$P < .05$ compared with the LOP group (ANOVA test).

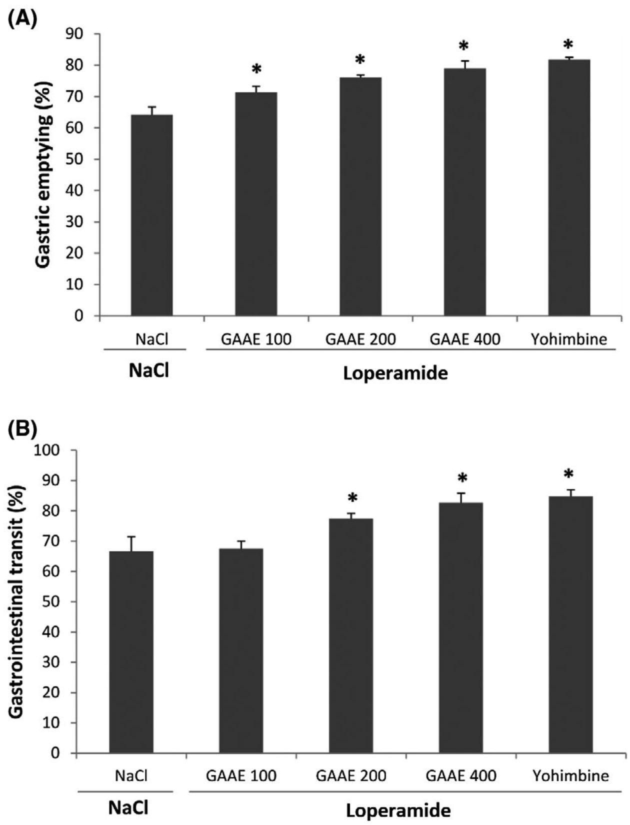

FIGURE 1 Acceleration of gastrointestinal motility by GAAE and yohimbine treatments. Animals were treated with various GAAE concentrations (100, 200, and 400 mg/kg, b.w. p.o.), or yohimbine (2 mg/kg, b.w., i.p.) or NaCl (0.9%). A, Effect on normal gastric emptying test meal (1.5 mL/rat) formed by 50 mg phenol red in 100 mL aqueous methyl cellulose (1.5%) was orally administrated after 1h, and animals were sacrificed after 30 min. B, Effect on normal intestinal transit. After 2 h of treatment, all rats received by gavage standard charcoal meal (10% charcoal in 5% gum Arabic, 1 mL/rat) and sacrificed after 30 min. Data are expressed as means ± SEM (n = 6). *P <.05 compared with the control group (ANOVA test)

2.10 | Statistical analysis

Data were subjected to one-way analysis of variance (ANOVA). All the values were expressed as mean ± standard error of the mean.[26] Values were considered statistically significant at $P < .05$.

3 | RESULTS

3.1 | GAAE opposes the LOP effect on food intake and water consumption

Food and water intake was measured every 24 hours. Rats that received only LOP showed a significant decrease in food intake (Table 2). This reduction signaled the induction of constipation in rats. However, rats pretreated with GAAE or with yohimbine consumed more food. Concerning water consumption, there was no significant variation among the different groups (Table 2).

3.2 | Antagonist effect of LOP and GAAE on fecal parameters

Total fecal pellets excreted by rats every day during the experiment were counted (Table 3). Results showed that rats treated by LOP had a fewer number of fecal pellets. This confirmed that they were constipated. However, pretreatment with GAAE increased defecation number. The same result was obtained with yohimbine treatment.

Water content was significantly reduced in LOP-treated rats. Defecations of GAAE-treated rats were richer in water, particularly with 400 mg/kg which was near to control ones (Table 3). Similarly, cinder which reflects mineral content in fecal pellets was more significant in GAAE-treated groups than in those that received LOP only (Table 3).

3.3 | GAAE accelerates gastric emptying

Compared with the negative control group, GAAE significantly increased gastric emptying in a dose-dependent manner (Figure 1A). The dose of 400 mg/kg accelerated phenol red meal to 81.72% which is comparable to the yohimbine one (85.21%).

3.4 | Increase of gastrointestinal transit under GAAE administration

Charcoal meal tests indicate the speed of food traversing throughout the gastrointestinal system and particularly peristaltic frequency. GAAE showed a significant increase of gastrointestinal transit with the 2 higher concentrations (Figure 1B). The higher concentration (400 mg/kg) showed a result very near to the reference molecular (yohimbine).

3.5 | LOP and GAAE modifying mucosal histology

After treatment colons were histologically examined to determine whether LOP could affect the histology of the mucosal layer. Sections stained with H & E in the normal colons showed a thick mucosal layer with a regular morphology (Figure 2(A) a). In contrast, treatment with LOP negatively affected the mucosal thickness. Surprisingly, LOP-treated colons showed inflamed mucosal characterized by the presence of edema and surface destruction (Figure 2(A) b). LOP and GAAE co-treatment enhanced the mucosal thickness. Additionally, GAAE reduced the edema and the inflammation in the colons with the two higher concentrations (Figure 2(A) d & e). The ones treated with yohimbine gave the same effect (Figure 2(A) f). Sections stained with Alcian blue were examined to evaluate the mucus-producing cell number which decreased under LOP treatment (Figure 2(B)). Co-treatment with GAAE partially restored the mucus-producing cell number (Figure 2(B) c, d & e). The same effect was found when colons were co-treated with yohimbine (Figure 2(B) f).

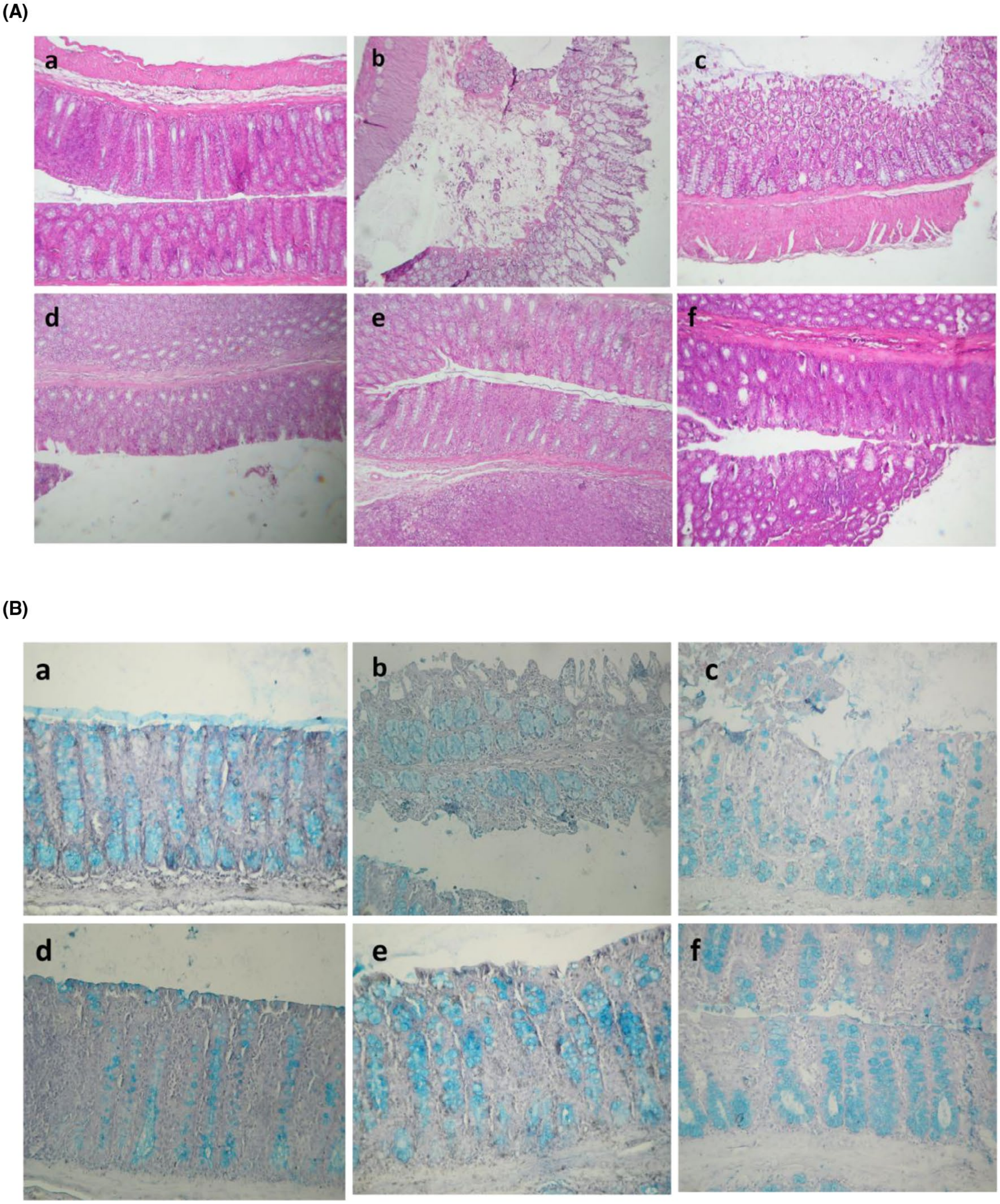

FIGURE 2 Effect of GAAE and Yohimbine on colon mucosa structure and layer thickness during LOP-induced constipation in rats. A, Hematoxylin and eosin staining (B) Alcian blue staining. Animals received LOP (3 mg/kg, b.w., i.p.) then treated with various GAAE concentrations (100, 200, and 400 mg/kg, b.w. p.o.), or yohimbine (2 mg/kg, b.w., i.p.) or NaCl (0.9%) during a week. (a) control, (b) LOP, (c) GAAE 100 mg/kg and LOP, (d) GAAE 200 mg/kg and LOP, (e) GAAE 400 mg/kg and LOP, and (f) yohimbine and LOP

3.6 | GAAE protects colon mucosa against LOP-lipoperoxydation

Lipoperoxydation was evaluated via MDA level measurement in colonic homogenate. Figure 5 illustrates MDA levels of all treatment groups. Colons treated with LOP showed a high MDA level when compared to untreated groups. This increase reflects damage in cell membrane exerted by LOP administration. Notwithstanding, groups treated with GAAE or with yohimbine showed lower MDA levels. The ones treated with 400 mg/kg of GAAE and with yohimbine showed levels near to the untreated ones (Figure 3).

3.7 | GAAE regenerates antioxidant enzymes after depletion under LOP administration

Superoxide dismutase (SOD) and gluthation peroxidase (GPx) enzymes levels showed depletion when treated only with LOP.

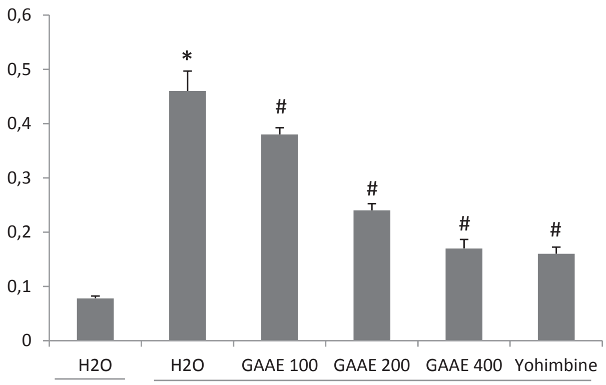

FIGURE 3 Effect of GAAE and yohimbine on colon lipoperoxydation (MDA nmol/mg protein) during LOP-induced constipation. Animals received LOP (3 mg/kg, b.w., i.p.) then treated with various GAAE concentrations (100, 200, and 400 mg/kg, b.w. p.o.), or yohimbine (2 mg/kg, b.w., i.p.) or NaCl (0.9%) over 1 wk. Data are expressed as means ± SEM (n = 10). *P <.05 compared with the control group and $^{\#}P$ < .05 compared with the LOP group (ANOVA test)

However, administration of yohimbine regenerates antioxidant enzymes levels. GAAE also significantly increased SOD and GPx levels in dose-dependent manner (Figure 4A,B).

3.8 | Effect of LOP and GAAE in Reactive oxygen species [8] levels

Figure 5 illustrates H_2O_2 (A) and OH^- (B) levels in colon homogenate. Notable production of the two ROS form was shown in LOP-treated colons (Figure 5A,B). This production decreased under treatment with yohimbine or GAAE. Significant effects with GAAE were shown with the two higher doses (200 and 400 mg/kg).

3.9 | Regeneration of-SH group levels by GAAE treatment

The -SH group level showed an important and significant decrease under LOP effect. However after treatment with yohimbine and GAAE, -SH groups were largely regenerated (Figure 6A).

3.10 | An overload in Ca^{2+} caused by LOP was followed by a down-regulation with GAAE administration

Variation of Ca^{2+} levels in different treated colons is represented in Figure 6B. We found that colons receiving LOP showed an overload in Ca^{2+} level. However, it seems that GAAE and yohimbine significantly down-regulated Ca^{2+} release in colon cells.

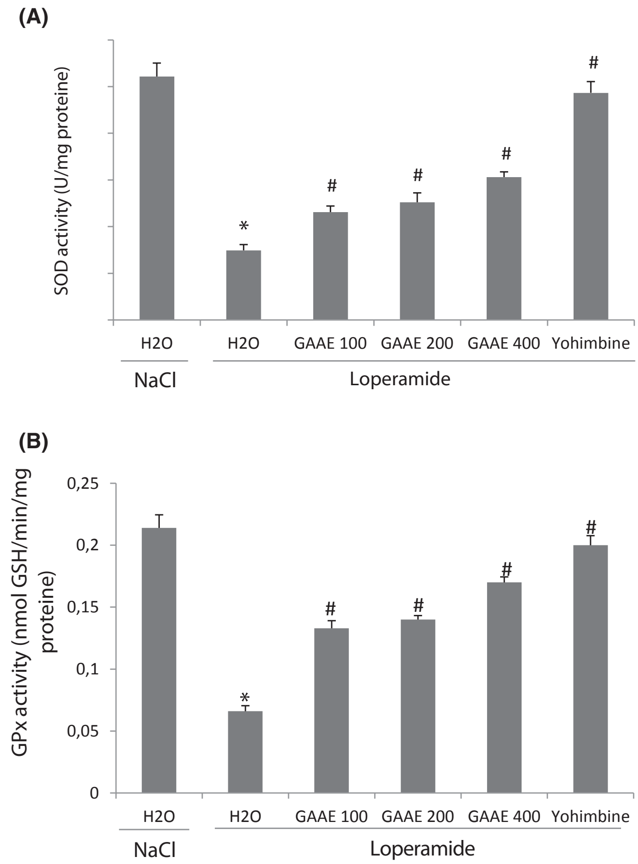

FIGURE 4 Colon antioxidant enzymes regeneration under GAAE and yohimbine treatment after depletion by LOP (A, SOD. U/mg protein; B, GPx. nmol GSH/min/mg protein). Animals received LOP (3 mg/kg, b.w., i.p.) then treated with various GAAE concentrations (100, 200, and 400 mg/kg, b.w. p.o.), or yohimbine (2 mg/kg, b.w., i.p.) or NaCl (0.9%) during a week. Data are expressed as means ± SEM (n = 10). *P < .05 compared with the control group and $^{\#}P$ < .05 compared with the LOP group (ANOVA test)

4 | DISCUSSION

Constipation is a common intestinal disorder that affects the life quality of patients.[1] Given the high cost and undesirable side effects of current drugs,[7] research is focused on investigation of herbals which have laxative and lubricating action. In a previous study, we determined the chemical composition of GAAE which was provided by flavonoids, iridoids, and phenylethanoid glucosides compounds.[10] Those secondary metabolites are known to exert several biological activities.[27-29] In this case, we investigated the purgative/laxative effects of GAAE on gastric emptying and GIT, and on constipation induced by LOP drug, testing the same doses (100, 200 and 400 mg/mL) used previously either in gastrointestinal motility or in colon inflammation which gave significant therapeutic properties with a minimum or any toxic side effects.[10,30] However, different equations should be used for dose extrapolation among Human or other species.[31] Therefore, the limits of the universal adequacy of herbal medicines due to the lack of standardized chemical characterization, the dosing

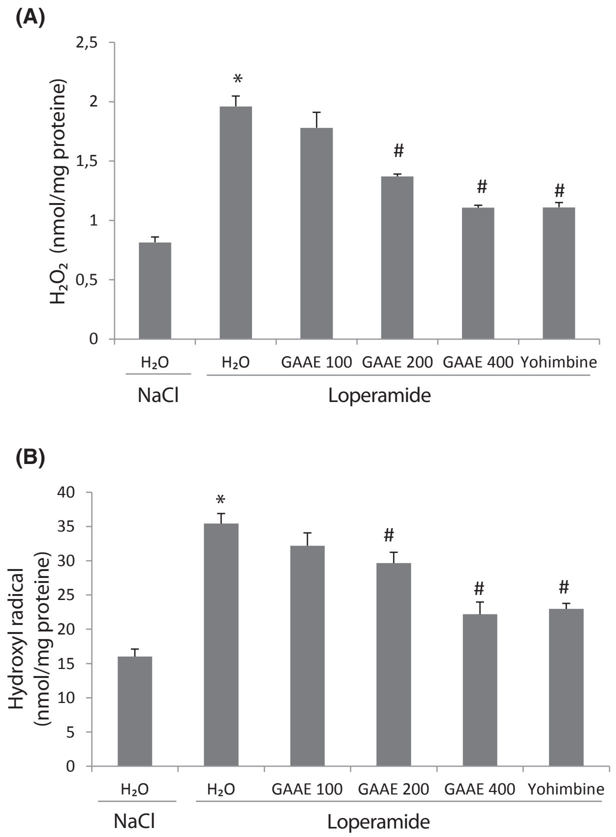

FIGURE 5 Effect of GAAE and yohimbine on colon ROS levels (A, H_2O_2 nmol/mg protein; B, hydroxyl radical nmol/mg protein) during LOP-induced constipation. Animals received LOP (3 mg/kg, b.w., i.p.) then treated with various GAAE concentrations (100, 200, and 400 mg/kg, b.w. p.o.), or yohimbine (2 mg/kg, b.w., i.p.) or NaCl (0.9%) during a week. Data are expressed as means ± SEM (n = 10). $^*P < .05$ compared with the control group and $^{\#}P < .05$ compared with the LOP group (ANOVA test)

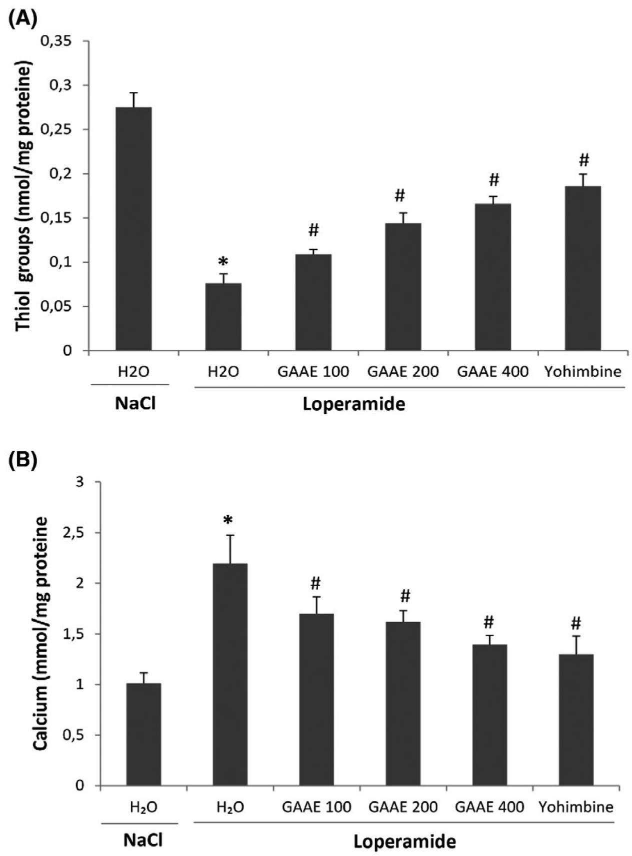

FIGURE 6 GAAE and yohimbine oppose LOP effect on thiol groups (A) (nmol/mg protein) and Ca^{2+} (B) (mmol/mg protein) during constipation. Animals received LOP (3 mg/kg, b.w., i.p.) then were treated with various GAAE concentrations (100, 200, and 400 mg/kg, b.w. p.o.), or yohimbine (2 mg/kg, b.w., i.p.) or NaCl (0.9%) over one week. Data are expressed as means ± SEM (n = 10). $^*P < .05$ compared with the control group and $^{\#}P < .05$ compared with the LOP group (ANOVA test)

schedule, and the substantial toxicity profile to verify their safety require intensive attention .[2]

Rats were treated with LOP and GAAE or yohimbine over seven days. The rats that received LOP consumed less food than the other groups. However, there was no change in water intake. The decrease in food intake has been documented in just a few other studies.[32,33] while the water intake result has been found in several studies.[32,34,35]

The number of fecal pellets as well as water and mineral content significantly decreased under LOP injection. This is a marker of induction of constipation which has been shown in most studies.[26,32,34,35] In fact, the drug inhibits intestinal water secretion [36]stimulates electrolyte absorption and inhibits stool frequency in humans.[37]

The administration of GAAE increased food intake as well as significantly increasing defecation number, water, and mineral content. Chinese medicines are shown as laxative and lubricating agents by stimulating Cl^- secretion and/or inhibiting sodium (Na^+) absorption across the colonic epithelium.[38] Flavonoids, one of the secondary metabolites present in GAAE, are suggested to activate a cAMP-dependent cystic fibrosis transmembrane conductance regulator (CFTR).[39] The CFTR is found in the apical membrane of epithelial cells[40] including the intestine[41] and could promote either reabsorption or secretion depending on the anion driving forces.[40]

The failure of peristalsis to move the luminal content through the colon is the major mechanism underlying constipation. That implies more time for salt and water absorption thereby reducing the frequency and weight of defecation.[42] The administration of GAAE significantly accelerated gastrointestinal transit with doses of 200 and 400 mg/kg. Intestinal contraction is modulated by smooth muscle cells which are innervated under acetylcholine releasing.[30] Our result correlates with the study of Fehri et al[30] which proved that GA increased intestinal peristalsis significantly at a dose of 400 mg/kg. In addition, the study showed that GA increased contraction of guinea pig isolated ileum by acetylcholine stimulate.[30]

GAAE improved gastric emptying rate especially with the dose of 400 mg/kg. This parameter determines nutrient release into the small intestine, which is particularly important for maintaining postprandial nutrient homeostasis.[43] Delayed gastric emptying can cause problematic upper GI symptoms.[44] In fact, gastric emptying is regulated by extrinsic innervation via vagal and splanchnic pathways while intrinsic innervation is modulated via the enteric nervous system.[45] In addition, several hormones influence gastric emptying such as ghrelin and motilin by their accelerative action.[46]

Added to that, the treatment for a week with LOP affected normal morphology of the colon layers. It not only disturbed the thickness of the mucosa, the crypts morphology, and the number of mucus-producing cells but also generated an unusual inflammation during the constipation induction. The effect of LOP in reducing mucosal thickness has been well documented in several previous studies[47-49] as well as the reducing in mucus-producing cell number.[50] In fact, the mucosal surface is covered by mucins, a hydrate gel produced in the intestine by goblet cells. This gel creates a barrier protecting the epithelium.[5] The decrease in mucus-producing cells contributes to defective mucus production which has been reported in numerous immune-mediated diseases.[51]

The treatment with LOP also caused an edema and inflammatory cell infiltration to the mucosa. This is the first study which signals the presence of inflammation in lop-induced constipation in rats. Only one recent study has reported the presence of colonic inflammation in mice.[52] The mucosal surface is implicated in the protection against inflammation process and the decrease in mucus-producing cells and the loss of epithelial integrity could generate mucosal inflammation.[6] The treatment with GAAE or yohimbine improved the epithelium integrity and increased the mucus-producing cells. Some previous studies had proved the antioxidant activity of GA and its richness in antioxidant compounds[53-55] in particular quercetin glucoside which protects the epithelium against physical pressure-associated damage.[56] GAAE also protected colons against inflammation induced by LOP. Data from a previous study have already showed the capacity of GAAE in protecting against colonic inflammation in rats.[10]

The oxidative status was evaluated in treated colons by measurement of some oxidant and antioxidant reactors. In this respect, it is showed that constipation induced by LOP in animals increase in turn the oxidative stress in the body that is exhibited with augmented levels of MDA, H_2O_2, and OH^- as well as depleted levels of enzymatic and non-enzymatic antioxidants such as SOD, GPx, and –SH groups. The constipation and antioxidant activity were significantly inversely correlated. This means that more constipation decreases the antioxidant activity of the body.[56] In addition, constipation provoked an accumulation of Ca^{2+} in colon cells. This massive increase of Ca^{2+} can generate more ROS production which reduces cell viability and, ultimately, leads to cell death. However, in our recent previous study, we clearly showed a positive correlation between laxative effects and antioxidant properties of medicinal plants such as *Malva sylvestri*[57] and Salvia officinalis.[58] In addition, taurine-xylose, a synthetic taurine-carbohydrate derivative, has also established a clear relationship between its antioxydant capacity and laxative effects in loperamide-induced constipation in Sprague Dawley rats.[59] Inversely, the treatment with GAAE or yohimbine reduced MDA, H_2O_2, OH^-, and Ca^{2+} levels and regenerated oxidant enzymes and –SH groups in a dose-dependent manner. In fact, the extract is rich in compounds[10] with antioxidant activity. Verbascoside, which is the major one, showed strong antioxidant capacity to inhibit lipid peroxidation in a membrane system model.[60] Moreover, a previous study found that catalpol, an iridoid compound of GAAE, [10] showed an antioxidant activity by increasing SOD and GPx activities and at the same time reduced MDA levels in liver and spleen.[61] Many other studies proved in vitro and in vivo antioxidant activity of GA extract.[27,55,62,63]

5 | CONCLUSION

In conclusion, many factors, for example, diet, malnutrition, metabolic disruptions, and side effects of medication, can lead to constipation and other gastrointestinal disorders. The results of recent research demonstrate that GAAE can exert a treatment action on constipation caused by the LOP drug. Indeed, GAAE exhibited laxative action without causing diarrhea and preventing ROS production. It is demonstrated that the supplementation with GAAE cured constipation and improved the antioxidant status of the body. Thus, GAAE may be effective as a complementary treatment for certain types of constipation in humans.

ACKNOWLEDGMENTS

Financial support of the Tunisian Ministry of Higher Education and Scientific Research is gratefully acknowledged. We would like to thank also Dr Maureen Galvin, Ph.D of the University of Salerno for the English revision.

CONFLICT OF INTEREST

The authors declare that they have no conflicting interests.

ETHICAL APPROVAL

All procedures on animals in this study were approved in conformity with the National Institute of Health recommendations for the use and care of animals.

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How to cite this article: Hajji N, Wannes D, Jabri M-A, et al. Purgative/laxative actions of *Globularia alypum* aqueous extract on gastrointestinal-physiological function and against loperamide-induced constipation coupled to oxidative stress and inflammation in rats. *Neurogastroenterol Motil*. 2020;32:e13858. https://doi.org/10.1111/nmo.13858